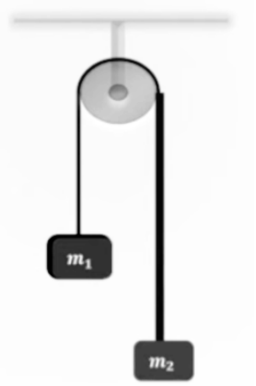
JEE-Main-09-04-2024 (Memory Based)
[MORNING SHIFT]

Physics

Question: The dimension of latent heat is:
Options:
(a) $[M^0L^2T^{-1}]$
(b) $[M^0L^2T^{-2}]$
(c) $[M^0LT^{-2}]$
(d) $[M^{-1}L^2T^{-2}]$
Answer: (b)

Question: In the pulley-block system shown, the pulley and the block are ideal. If the acceleration of the block is g/8, find $m_1 : m_2$ (Given $m_2 > m_1$)



Options:
(a) 7 : 9
(b) 5 : 7
(c) 3 : 4
(d) 9 : 11
Answer: (a)

Question: Velocity of a particle of mass m as a function of displacement x is given by $v = \alpha\sqrt{x}$. Work done to move it from x = 0 to x = d is:
Options:
(a) $\frac{m\alpha^2}{2} \cdot d$
(b) $m\alpha^2 \cdot d$
(c) $3m\alpha^2 \cdot \frac{d}{2}$
(d) $2m\alpha^2 d$
Answer: (a)

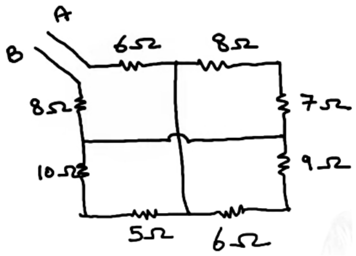
Question: Two persons are pulling a rope towards themselves with a force of 200 N each. If the Young's modulus is 2×10^{11} N/m^2 and area of cross-section is 2 cm^2 for the rope, the elongation in the rope is __
(distance between the persons holding the ropes is 2 m)
Options:
(a) 10 μm
(b) 20 μm
(c) 5 μm
(d) 40 μm
Answer: (a)

Question: A particle oscillating in simple harmonic motion such that its speed and acceleration at distance 2 m from mean position are 4 m/s and 16m/s^2 respectively. Find the amplitude of oscillation of the particle.
Options:
(a) $\sqrt{10}$ m
(b) $\sqrt{6}$ m
(c) $\sqrt{8}$ m
(d) $\sqrt{3}$ m
Answer: (b)

Question: Assertion (A): Object at radius of curvature of biconvex lens made by glass (μ = 1.5) form image at same distance an other side of the lens.
Reason (R): Image of a real object formed by concave lens is always virtual and erect.
Options:
(a) Both Assertion(A) and Reason(R) are the true and Reason(R) is a correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are the true but Reason(R) is a correct explanation of Assertion (A)
(c) Aseration (A) is true and Reason (R) is false
(d) Assertion (A) is false and Reason (R) is true
Answer: (b)

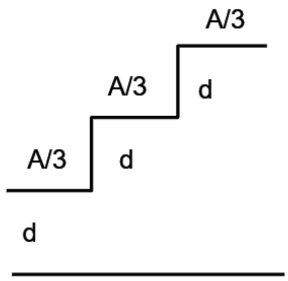
Question: The equivalent energy of 1 gm mass is equal to:
Options:
(a) 8.3×10^{26} M_eV
(b) 5.6×10^{26} M_eV
(c) 8.3×10^{12} M_eV
(d) 5.6×10^{12} M_eV
Answer: (b)

Question: Find the equivalent resistance between terminal A and B for the given network.



Options:
(a) 16 Ω
(b) 20 Ω
(c) 15 Ω
(d) 19 Ω
Answer: (d)

Question: Find the Equivalent capacitance from given diagram



Options:
(a) $11A\varepsilon_o/18d$
(b) $13A\varepsilon_o/18d$
(c) $15A\varepsilon_o/18d$
(d) $18A\varepsilon_o/11d$
Answer: (a)

Question: A monatomic gas is expanded adiabatically from 4V to 5V find ratio of initial and final Pressure?
Options:
(a) $5\sqrt{5}/8$
(b) $5\sqrt{7}/8$
(c) $5\sqrt{5}/7$
(d) $7\sqrt{5}/8$
Answer: (a)

Question: A gas is expanded adiabatically - initial temperature is T volume V. The final volume 2V then find work done in process
Options:
(a) $RT(2-\sqrt{2})$
(b) $R/T(2-\sqrt{2})$
(c) 2RT

(d) 5RT
Answer: (a)

Question: A half ring of radius R = 10 cm has linear charge density 4nC/m. Its potential at the centre is given as $x\pi$ V. Find x = ?
Options:
Answer: (36)

Question: Intensity at point is ¼ of max intensity. find its minimum distance from centre. Given wavelength 600 nm, d = 1mm, D = 1m
Options:
Answer: (200 μm)

Question: An astronaut takes a ball of mass m from earth surface. He throws the ball in the radius of 386.6. Then change in potential energy is xGMm/21R Find x.
Take radius of earth R = 6310 km.
Options:
Answer: (11)

Question: Rod weight W kept on shoulder of man inclined at angle θ find weight experienced
Options:
(a) W
(b) W/2
(c) W/4
(d) W/8
Answer: (b)

Question: Find the Ratio of De broglie wavelength of α - article, Electron & Proton
Options:
(a) $\lambda_\alpha < \lambda_p < \lambda_e$
(b) $\lambda_\alpha > \lambda_p > \lambda_e$
(c) $\lambda_\alpha > \lambda_p < \lambda_e$
(d) $\lambda_\alpha < \lambda_p > \lambda_e$
Answer: (a)

Question: If a bulb and capacitor are connected in series and then capacitor is inserted with a dielectric then
Bulb's intensity
Options:
(a) Increases
(b) Decreases
(c) Remains same
(d) Becomes 0
Answer: (a)

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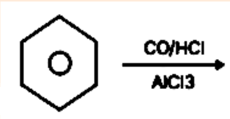
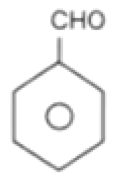
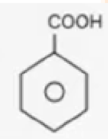
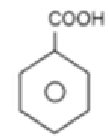
Chemistry

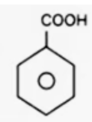
Question: Which will not form when PbS react with dil HNO_3
Options:
(a) NO
(b) NO_2
(c) S
(d) None
Answer: (b)

Question: For the reaction : 

Product (P) is :
Options:
(a) 

(b) 

(c) 

(d) 

Answer: (a)

Question: Which of the following has sp^2 hybridisation ?
Options:
(a) BF_3
(b) H_2SO_4
(c) NH_4^+
(d) NH_3
Answer: (a)

Question: Consider the following electronic configuration :

$Cu^{2+} = [Ar]3d^4s^0$

$Cu^+ = [Ar]3d^{10}4s^0$

Which option is correct ?

Options:

(a) Cu^{2+} is more stable in aqueous solution

(b) Cu^+ is more stable in aqueous solution

(c) Cu^{2+} and Cu^{2+} are equally stable in aqueous solution

(d) Depends upon copper salt

Answer: (a)

Question: Chemical formula of compound present in tooth enamel ?

Options:

(a) $Ca_{10}(PO_4)_6(OH)_2$

(b) $Ca_8(PO_4)(OH)_2$

(c) $Ca_6(PO_4)_2(OH)_2$

(d) $Ca_8(PO_4)_6(OH)_2$

Answer: (a)

Question: Equal volume of 1 M HCl and 1 M H_2SO_4 neutralized by dil. NaOH and heat released is x and y kcal respectively, then which is correct ?

Options:

(a) x= y

(b) x = 0.5 y

(c) x = 0.4 y

(d) x = 2y

Answer: (b)

Question: Number of ambidentate nucleophiles among the following is CN^-, SCN^- ,NO^-_2 , CH_3COO^-, NH^-_2 , SO^{2-}_4 :

Options:

Answer: (3)

Question:Which of the following orbitals has the highest energy ?

Options:

(a) n = 6, l = 0

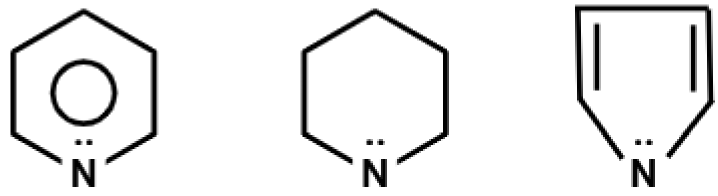
(b) n = 5, l = 2

(c) n = 4, l = 2

(d) n = 3, l = 1

Answer: (b)

Question: Arrange the following compounds on the Basis of basic strengths

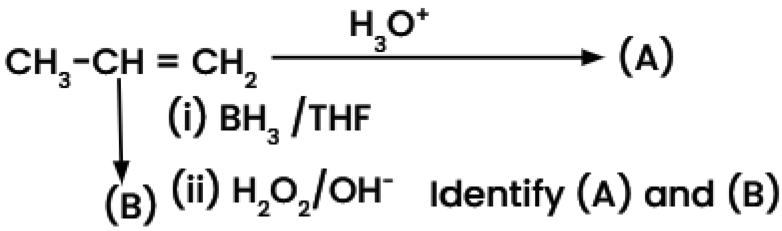


(a) (b) (c)

Options:
(a) a > b > c
(b) b > a > c
(c) c > a > b
(d) b > c > a
Answer: (a)

Question: Find number of Geometrical Isomer in $[M(AH)_2a_2]$
Options:
(a) 4
(b) 3
(c) 2
(d) 1
Answer: (c)

Question:



Options:
(a) A → CH_3 - CH - CH_3 B → CH_3 - CH_2 - CH_2 - OH

|

OH

(b) A → $CH_3\ CH_2\ CH_2$-OH   B→ CH_3 - CH - CH_3

|

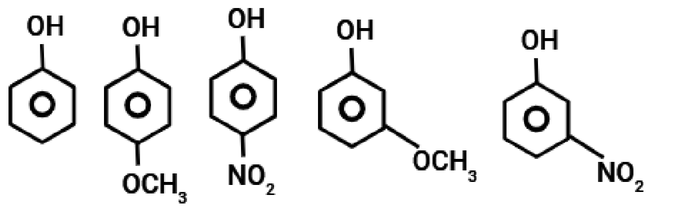
OH

(c) A → $CH_3\ CH_2\ CH_3$ B→ $CH_3CH_2CH_2$-OH

(d) A → $CH_3\ CH_2\ CH_2\ OH$ B→ $CH_3CH_2CH_3$

Answer: (a)

Question: Arrange the following in increasing order of acidity.



Options:

(a) I < II < III < IV < V

(b) II < I < IV < V < III

(c) III < V < IV < I < II

(d) II < IV < III < I < V

Answer: (b)

Question: Which of the following is colorless ?

Options:

(a) Eu^{3+}

(b) Lu^{3+}

(c) Nd^{3+}

(d) Sm^{3+}

Answer: (b)

Question: Which among the following have single unpaired electron ?

N_2 , O_2 , CN^- , O^-_2 , C^{2-}_2 , N^-_2

Options:

(a) O_2 , N_2

(b) CN^- , C^{2-}_2

(c) CN^- , O^-_2

(d) N^-_2 , O^-_2

Answer: (d)

Question: Statement - I : Sulphur exists as S_8 while oxygen exists as O_2.

Statement - II : In oxygen , pπ - pπ bonding occurs while it is not effective in sulphur.

Options:

(a) Both S- I and S - II are true

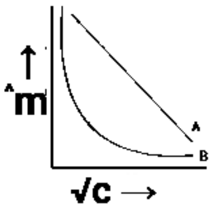
(b) S- I is true and S - II is false

(c) S- I is false and S - II is true
(d) Both S- I and S - II are false
Answer: (a)

Question: Which of the following statement is incorrect ?
Options:
(a) $KMnO_4$ and NaOH can be used as secondary standard
(b) Primary standard should not undergo change in air
(c) Reaction of primary standard with another substance should not be instantaneous
(d) Primary standard should be soluble in H2O
Answer: (c)

Question: Purification method of organic compound does not depend on :
Options:
(a) Nature of compound
(b) Shape of compound
(c) Density of compound
(d) Solubility of compound
Answer: (b)

Question: Molar conductance vs √concentration curve for two electrolytes 'A' and 'B' are shown. Identify the nature of both electrolytes :



Options:
(a) A → Strong electrolyte ; B → Strong electrolyte
(b) A → Weak electrolyte ; B → Strong electrolyte
(c) A → Strong electrolyte ; B → Weak electrolyte
(d) A → Weak electrolyte ; B → Weak electrolyte
Answer: (c)

Question: Total number of essential amino acids are :
Options:

(a) 12
(b) 11
(c) 10
(d) 9
Answer: (c)

Question: Rate of a reaction is given as rate = $k[A]^2[B]$. If concentration of both reactants is doubled then rate becomes x times the previous and the order of reaction is y, then what is the value of (x + y)
Options:
Answer: (11)

Question: Consider the given complex, $[Co(en)_2Cl_2]^+$.
Statement - I :- The number of stereoisomers for the above compound is 3.
Statement - II :- Geometry of the above complex is octahedral
Options:
(a) S- I is correct, S - II is incorrect
(b) S- I is correct, S - II is correct
(c) S- I is incorrect and S - II is correct
(d) S- I is incorrect, S - II is incorrect
Answer: (b)

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Maths

Question: A ray of light passing through (1,2) after reflecting on x- axis at point Q passes through R(3, 4). If S(h,k) is such that PQRS is a parallelogram, then find the value of hk^2.
Options:
(a) 90
(b) 84
(c) 98
(d) 108
Answer: (b)

Question: Tetrahedral dice having outcomes (1,2,3,4)
Options:
(a) 4
(b) 5
(c) 6
(d) 7
Answer: (b)

Question: A circle passes through (0,0) and (1,0) and touches the circle $x^2 + y^2 = 9$. Then the locus of the centre of the circle is:
Options:
(a) Circle
(b) Parabola
(c) Hyperbola
(d) Straight Line
Answer: (a)

Question: $\vec{A}, \vec{B}$ and $\vec{C}$ are given as $\vec{A} = \alpha\hat{i} + 4\hat{j} + 5\hat{k}$ $\vec{B} = 2\hat{i} + 5\hat{j} + 6\hat{k}$ $\vec{C} = \vec{A} + \vec{B}$ $\left|\vec{C}\right| = \left|\vec{A} - \vec{B}\right|$ is equal to:
Options:
(a) 25,731
(b) -25,669
(c) -25,731
(d) 25,669
Answer: (c)

Question: If set A = $\{z : |z-1| \leq 1\}$ and set B = $\{z : |z-5i| \leq |z-5|\}$, if $z = a + ib$, where $a, b \in \mathbb{I}$, then sum of modulus squares of $A \cap B$ is :
Options:

(a) 0
(b) 2
(c) 4
(d) 5
Answer: (b)

Question: If $\frac{1}{(1+d)(1+2d)} + \frac{1}{(1+2d)(1+3d)} + ... + \frac{1}{(1+9d)(1+10d)} = 1,$ then value of 50d is (d > 0):
Options:
(a) 50
(b) 60
(c) 25
(d) 30
Answer: (c)

Question: The remainder when $(428)^{2024}$ is divided by 21 is:
Answer: (1)

Question: If A is 3 x 3 matrix, det(3adj (2 adj A)) = 2^{-13}. 3^{-10} and det(3adj (2A)) =2^{-m}. 3^{-n} then $2^m + 2^n$ is equal to :
Answer: (14)

Question: If f(x) =$3ax^3 + bx^2 + cx + 1$ and f(1) = 41, f'(1) = 2 and f''(1) = 4 then $(a^2 + b^2 + c^2)$ is ____.
Answer: (8)

Question: If domain of f(x) $\sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is R - $(\alpha, \beta]$ then $12\alpha\beta$ is equal to :
Answer: (32)

Question: A = {2,4,6,8}, B = {3,7,6,9}. R : A X B → A X B such that (a_1, b_1) R (a_2, b_2) ⇔ $a_1 + a_2 = b_1 + b_2$ where $(a_1, b_1) \in A$, $(a_2, b_2) \in B$. Find number of elements in the relation.
Answer: (9)

Question: Let $\int \frac{2 - \tan x}{3 + \tan x} dx = \frac{1}{2}(ax + \log_e|\beta \sin x + v \cos x|) + C,$ where c is constant of integration. Then $\alpha + \frac{\gamma}{\beta} =$
Options:
(a) 4
(b) 7
(c) 1
(d) 3
Answer: ()

Question: Let $|\cos\theta \cos(60 - \theta)| \leq \frac{1}{8}$, $\theta \in [0, 2\pi]$. Then the sum of all $\theta \in [0, 2\pi]$, where $\cos 3\theta$ attains its maximum value is

Options:
(a) 15π
(b) 9π
(c) 6π
(d) 18π
Answer: (c)

Question: The coefficient of x^{70} in $x^2(1 + x)^{98} + x^3(1 + x)^{97} + x^4(1 + x)^{96} + ... + x^{54}(1 + x)$ ^9C_p $-^{46}C_q$ is. Then a possible value of p + q is
Options:
(a) 68
(b) 55
(c) 83
(d) 61
Answer: (c)

Question: A variable line L passes through the point (3,5), and intersects the +ve coordinate axes at the points A and B. The minimum area of the ΔOAB, where o is the origin is :
Options:
(a) 25
(b) 40
(c) 35
(d) 30
Answer: (d)